Cyclohexane Reactor Ver 1 · operator training simulator

Line list and line sizing

Document no. CHX-V1-PE-LL-001 · **Discipline** Process engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 40 main lines: 14 process liquid (1001…), 12 process gas and hydrogen (2001…), 14 utility (3001…).
• Line number **<NPS>"-<service>-<sequence>-<spec>**, e.g. 3"-P-1002-B1; the same numbers go on the P&ID CHX-V1-PID-001.
• Design flows from the heat and material balance CHX-V1-PE-HMB-001 (largest of the design, 120 % and 70 % cases) or the model's own maximum for pumps and utilities.
• Every selected size meets its velocity / ΔP / ρv² / Mach criteria at the design flow (40 of 40); 3 slurry lines fall below their minimum velocity at the lowest case (EQ-ep-13).
• Design pressures and temperatures from the Safety-layers assumptions (loop 36 bar g, K-101 discharge 42 bar g, R-101 250 °C, R-103 260 °C) or labelled assumed. 13 engineering queries raised.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Line-numbering scheme

Format **<NPS>"-<service>-<sequence>-<spec>**. NPS in inches (1-1/2 for 1½"). The sequence number is unique across the unit and does not change when the size changes; a spec break (for example at a control valve) starts a new line number.

Service code	Meaning	Lines
P	process liquid (incl. catalyst slurry)	14
PG	process gas and gas–liquid	6
H	hydrogen (make-up, recycle, purge)	6
S	steam (LP, MP; steam–water riser)	4
BFW	boiler feed water and drum water	2
CW	cooling water	5
N	nitrogen	1
FL	flare and vents	2
1001–1999	sequence numbers: process liquid	
2001–2999	sequence numbers: process gas and hydrogen	
3001–3999	sequence numbers: utilities	

Piping specs (assumed, typical)

Spec	Class	CA mm	Description
A1	150	1.5	CS, ASME class 150, CA 1.5 mm — utilities (water, steam ≤ 12 bar g, N_2)
A2	150	3.0	CS, class 150, CA 3 mm — low-pressure hydrocarbon (product to tanks, fuel gas, flare)
B1	300	3.0	CS, class 300, CA 3 mm — hydrocarbon liquid and gas
B2	300	3.0	CS (killed, PWHT), class 300, CA 3 mm — hydrogen-rich gas; API 941 check for hot H_2
B3	300	3.0	CS, class 300, CA 3 mm + erosion allowance — catalyst slurry: long-radius bends, no dead legs, flushing connections

2 · Sizing basis

Design flow = the largest of the three HMB cases (the rated case), except: P-101 lines at the pump's model maximum K.Fpa (the three-way FV-104 only splits a constant pump flow); utility lines at the model's own maximum (valve wide open). **Lowest** = the smallest non-zero case flow, used for the minimum velocity of catalyst-slurry and water lines. **Size** = the smallest nominal size meeting every maximum criterion; schedule 80 up to 1½", 40 above, raised to 80 where the ASME B31.3 thickness needs it (S = 130 MPa assumed, E = W = 1, Y = 0.4, mill tolerance 12.5 %, CA per spec). Friction: Darcy, Colebrook (iterated), roughness 0.046 mm. Properties: liquid density from the model (benzene typical), gas ideal, two-phase homogeneous; viscosities typical. Equivalent lengths are assumed (layout not fixed).

Code	Service	Criteria (typical practice)
L-suc	Liquid, pump suction	v ≤ 1.0 m/s, ΔP ≤ 0.25 bar/100 m
L-dis	Liquid, pump discharge	v 1–3 m/s, ΔP ≤ 0.9 bar/100 m
L-cv	Liquid, level-valve and transfer lines	v ≤ 2.5 m/s, ΔP ≤ 0.6 bar/100 m

Code	Service	Criteria (typical practice)
L-grav	Liquid, gravity (reflux)	$v \le 1.0$ m/s, $\Delta P \le 0.15$ bar/100 m
SL-suc	Catalyst slurry, pump suction	v 1.0–2.0 m/s (solids kept moving), $\Delta P \le 0.4$ bar/100 m
SL	Catalyst slurry	v 1.0–3.0 m/s (settling / erosion), $\Delta P \le 1.0$ bar/100 m
G	Process gas	$\rho v^2 \le 6\,000$ Pa, $v \le 30$ m/s, $\Delta P \le 0.5$ bar/100 m
G-loop	Recycle-loop gas	$\rho v^2 \le 6\,000$ Pa, $\Delta P \le 0.2$ bar/100 m (the whole loop has ≈ 3 bar from K-101)
G-suc	Gas, compressor suction	$\rho v^2 \le 6\,000$ Pa, $v \le 25$ m/s, $\Delta P \le 0.2$ bar/100 m
G-dis	Gas, compressor discharge	$\rho v^2 \le 15\,000$ Pa, $v \le 30$ m/s, $\Delta P \le 0.5$ bar/100 m
2P	Two-phase gas–liquid	$\rho_m v^2 \le 10\,000$ Pa (erosion), $v \le 20$ m/s, $\Delta P \le 0.6$ bar/100 m
2P-loop	Two-phase, recycle loop	$\rho_m v^2 \le 10\,000$ Pa, $\Delta P \le 0.3$ bar/100 m (loop budget)
ST	Steam, LP and MP	$v \le 35$ m/s, $\Delta P \le 0.25$ bar/100 m
W	Water (CW, BFW)	v 1–3 m/s, $\Delta P \le 0.6$ bar/100 m
W-dc	Saturated water, downcomer	$v \le 1.5$ m/s, $\Delta P \le 0.15$ bar/100 m
N	Utility gas (N_2)	$v \le 25$ m/s, $\Delta P \le 0.5$ bar/100 m
FL	Flare / vent	Mach ≤ 0.5, $\rho v^2 \le 100\,000$ Pa

Design pressure: the set of the relief device protecting the line (Safety-layers SL_ASSUMED) or the pump shut-off, whichever governs; **design temperature**: equipment design temperature for lines on R-101/R-103 and for condenser outlets (loss of cooling), otherwise maximum operating + 30 °C rounded up (typical). The source is given for every line in table 3b.

3 · Line list

3a · Process data and sizing

Line number	From → to	PFD	Fluid (phase)	Design flow t/h	Lowest t/h	ρ kg/m³	μ cP	Criteria	NPS / sch	ID mm	v m/s	Re	ΔP bar /100 m	ρv² Pa	Mach	v lowest m/s	OK
6"-P-1001-B1	TK-100 → P-100A/B suction	1	benzene (L)	41.8	24.4	858	0.50	L-suc	6" / 40	154.1	0.73	193 k	0.026	451	—	0.42	✓
3"-P-1002-B1	P-100A/B → R-101 (FV-101, XV-101)	2	benzene (L)	41.8	24.4	858	0.50	L-dis	3" / 40	77.9	2.84	381 k	0.816	6,898	—	1.65	✓
18"-P-1003-B3	R-101 bottom → P-101A/B suction	9	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	605.5	604.7	607	0.15	SL-suc	18" / 40	428.7	1.92	3.33 M	0.033	2,239	—	1.91	✓
16"-P-1004-B3	P-101A/B discharge → FV-104 (three-way)	10	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	605.5	604.7	607	0.15	SL	16" / 40	381.0	2.43	3.75 M	0.060	3,587	—	2.42	✓
16"-P-1005-B3	FV-104 → E-101 tube inlet	10	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	605.5	224.2	607	0.15	SL	16" / 40	381.0	2.43	3.75 M	0.060	3,587	—	0.90 !	✓
16"-P-1006-B3	E-101 outlet (+ by-pass) → R-101	11	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	605.5	604.7	607	0.15	SL	16" / 40	381.0	2.43	3.75 M	0.060	3,587	—	2.42	✓
16"-P-1007-B3	FV-104 by-pass → R-101 return	10B	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	605.5	151.9	607	0.15	SL	16" / 40	381.0	2.43	3.75 M	0.060	3,587	—	0.61 !	✓
4"-P-1008-B3	R-101 (LV-101) → R-102	8	cyclohexane + 2.0 mol % benzene + Raney-Ni slurry (L)	43.6	25.7	608	0.15	SL	4" / 40	102.3	2.43	1.01 M	0.295	3,585	—	1.43	✓
8"-P-1009-B1	V-103 → R-101 (reflux)	6	cyclohexane + 2.1 mol % benzene (L)	55.1	33.1	733	0.55	L-grav	8" / 40	202.7	0.65	176 k	0.013	306	—	0.39	✓
4"-P-1010-B3	R-102 bottom → P-102A/B suction	13	cyclohexane + Raney-Ni slurry (L)	25.3	14.5	641	0.20	SL-suc	4" / 40	102.3	1.34	440 k	0.098	1,145	—	0.77 !	✓
3"-P-1011-B3	P-102A/B → F-102A/B	14	cyclohexane + Raney-Ni slurry (L)	25.3	14.5	641	0.20	SL	3" / 40	77.9	2.30	578 k	0.394	3,397	—	1.32	✓
3"-P-1012-B1	F-102A/B (LV-102) → R-103	15	cyclohexane (L)	25.3	14.5	641	0.20	L-cv	3" / 40	77.9	2.30	578 k	0.394	3,397	—	1.32	✓
4"-P-1013-B1	V-104 → LV-104	21	cyclohexane (L)	45.0	26.2	743	0.67	L-cv	4" / 40	102.3	2.05	234 k	0.279	3,117	—	1.19	✓
4"-P-1014-A2	LV-104 (HS-115) → TK-106 / TK-107	21	cyclohexane (L)	45.0	26.2	743	0.67	L-cv	4" / 40	102.3	2.05	234 k	0.279	3,117	—	1.19	✓
10"-PG-2001-B2	R-101 overhead → E-103	4	H_2 + CH_4 + C_6 vapour (V)	61.4	35.8	31.2	0.011	G-loop	10" / 40	254.6	10.75	8.12 M	0.096	3,604	—	6.42	✓
8"-PG-2002-B2	E-103 outlet → V-103	5	H_2-rich gas + cyclohexane liquid (V/L)	61.4	35.8	69.6	0.087	2P-loop	8" / 40	202.7	7.59	1.23 M	0.146	4,008	—	4.33	✓

Line number	From → to	PFD	Fluid (phase)	Design flow t/h	Lowest t/h	ρ kg/m³	μ cP	Criteria	NPS / sch	ID mm	v m/s	Re	ΔP bar /100 m	ρv² Pa	Mach	v lowest m/s	OK
6"-PG-2003-B2	V-103 overhead → R-102 sparger	7	H_2 76.9 %, CH_4 21.6 % (V)	6.33	2.7	7.71	0.011	G-loop	6" / 40	154.1	12.24	1.38 M	0.058	1,155	—	6.87	✓
8"-PG-2004-B2	R-102 overhead → R-103 inlet	12	H_2 + CH_4 + C_6 vapour (V)	24.6	13.9	19.3	0.011	G-loop	8" / 40	202.7	10.97	3.98 M	0.082	2,325	—	6.55	✓
8"-PG-2005-B2	R-103 outlet → E-106	16	H_2-rich gas + cyclohexane liquid (V/L)	50.0	28.4	37.6	0.021	2P-loop	8" / 40	202.7	11.42	4.17 M	0.174	4,913	—	6.72	✓
8"-PG-2006-B2	E-106 outlet → V-104	17	H_2-rich gas + cyclohexane liquid (V/L)	50.0	28.4	60.6	0.080	2P-loop	8" / 40	202.7	7.09	1.09 M	0.112	3,051	—	4.14	✓
8"-H-2007-B2	V-104 overhead → K-101 suction	18	H_2 92.1 %, CH_4 7.9 % (V)	8.22	4.0	3.79	0.009 8	G-suc	8" / 40	202.7	18.67	1.46 M	0.048	1,320	—	10.89	✓
8"-H-2008-B2	K-101 discharge → FV-103	3	H_2 92.1 %, CH_4 7.9 % (V)	8.22	4.0	3.89	0.010	G-dis	8" / 40	202.7	18.17	1.41 M	0.047	1,285	—	10.12	✓
8"-H-2009-B2	FV-103 → R-101 sparger	3	H_2 92.1 %, CH_4 7.9 % (V)	8.22	4.0	3.89	0.010	G-loop	8" / 40	202.7	18.17	1.41 M	0.047	1,285	—	10.12	✓
6"-H-2010-B1	Make-up H_2 battery limit (XV-105, PV-105) → V-104	19	H_2 99.5 %, CH_4 0.5 % (V)	3.53	2.1	2.88	0.009 2	G	6" / 40	154.1	18.25	877 k	0.049	961	—	11.05	✓
1-1/2"-H-2011-B2	V-104 → HIC-107	20	H_2 92.1 %, CH_4 7.9 % (V)	0.30	0.3	3.79	0.009 8	G	1-1/2" / 80	38.1	19.27	283 k	0.396	1,407	—	19.27	✓
3"-H-2012-A2	HIC-107 → Fuel-gas battery limit	20	H_2 92.1 %, CH_4 7.9 % (V)	0.30	0.3	0.757	0.009 8	G	3" / 40	77.9	23.10	139 k	0.052	404	—	23.10	✓
10"-S-3001-A1	D-101 (PV-110) → LP steam header	22	LP steam (V)	18.0	9.6	2.93	0.014	ST	10" / 40	254.6	33.56	1.73 M	0.091	3,300	—	17.88	✓
8"-S-3002-A1	MP steam header (HV-110) → E-101 shell (HS-101 HEAT)	—	MP steam (V)	15.1	—	5.49	0.016	ST	8" / 40	202.7	23.70	1.69 M	0.111	3,083	—	—	✓
6"-S-3003-A1	MP steam header (TV-102) → R-102 jacket (HS-102 HEAT)	—	MP steam (V)	4.66	—	5.49	0.016	ST	6" / 40	154.1	12.63	685 k	0.045	876	—	—	✓
3"-BFW-3004-A1	BFW header (LV-109) → D-101	23	boiler feed water (L)	18.2	—	954	0.27	L-cv	3" / 40	77.9	1.11	312 k	0.141	1,179	—	—	✓
8"-BFW-3005-A1	D-101 downcomer → E-101 shell inlet	—	saturated water (L)	144.2	—	903	0.17	W-dc	8" / 40	202.7	1.37	1.45 M	0.062	1,704	—	—	✓
16"-S-3006-A1	E-101 shell outlet (riser) → D-101	—	steam–water (V/L)	144.2	—	22.9	0.073	2P	16" / 40	381.0	15.32	1.83 M	0.093	5,381	—	—	✓
10"-CW-3007-A1	CW supply (HV-113) → E-103	—	cooling water (L)	432.0	—	994	0.76	W	10" / 40	254.6	2.37	785 k	0.161	5,593	—	—	✓
10"-CW-3008-A1	E-103 → CW return	—	cooling water (L)	432.0	—	987	0.54	W	10" / 40	254.6	2.39	1.12 M	0.159	5,631	—	—	✓
8"-CW-3009-A1	CW supply → E-106	—	cooling water (L)	216.0	—	994	0.76	W	8" / 40	202.7	1.87	493 k	0.134	3,475	—	—	✓

Line number	From → to	PFD	Fluid (phase)	Design flow t/h	Lowest t/h	ρ kg/m³	μ cP	Criteria	NPS / sch	ID mm	v m/s	Re	ΔP bar /100 m	ρv² Pa	Mach	v lowest m/s	OK
8"-CW-3010-A1	E-106 → CW return	—	cooling water (L)	216.0	—	988	0.56	W	8" / 40	202.7	1.88	677 k	0.132	3,496	—	—	✓
6"-CW-3011-A1	CW supply (TV-102) → R-102 jacket (HS-102 COOL)	—	cooling water / MP steam (L)	144.0	—	994	0.76	W	6" / 40	154.1	2.16	433 k	0.247	4,630	—	—	✓
16"-N-3012-A1	N_2 header 8 bar g → HV-120 → loop	—	nitrogen (V)	99.8	—	10.0	0.018	N	16" / 40	381.0	24.30	5.14 M	0.099	5,905	—	—	✓
8"-FL-3013-A2	EDV-101 outlet → Flare header	—	loop gas to flare (V)	8.09	—	0.177	0.0098	FL	8" / 40	202.7	392.29	1.45 M	0.991	27,295	0.36	—	✓
12"-FL-3014-A2	HV-121 vent → Flare header	—	loop gas to flare (V)	20.2	—	0.177	0.0098	FL	12" / 40	303.3	438.27	2.42 M	0.759	34,068	0.40	—	✓

Criteria codes: see section 2. ! = below the minimum velocity of its criteria at the lowest case (EQ-ep-13). OK = every maximum criterion met at the design flow.

3b · Design conditions, spec and notes

Line number	P op / des bar g	Design pressure source	T op / des °C	Design temperature source	Spec	Ins.	Le m	Design-flow basis and notes
6"-P-1001-B1	1.0 / 40.0	suction rated as the discharge up to the suction block valve (spare-pump back-flow), typical	40 / 70	max operating 40 °C + 30 °C, rounded up (typical)	B1	—	60	largest of the three HMB cases. Benzene freezes at 5.5 °C: no tracing needed at site ambient (assumed ≥ 20 °C).
3"-P-1002-B1	30.4 / 40.0	P-100 shut-off 35.0 bar g (model: no feed when PI-101 ≥ 36 bar a) + suction 1 bar g, rounded up (assumed)	40 / 70	max operating 40 °C + 30 °C, rounded up (typical)	B1	—	80	largest of the three HMB cases. Spec break B1 holds the shut-off pressure; check valve NRV-100 at each pump (Safety layers SL-R21).
18"-P-1003-B3	29.5 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	20	P-101 rated flow = model maximum K.Fpa (constant: the three-way FV-104 only splits it). R-101 liquid is at its bubble point with dissolved H_2: NPSH from static head only (part 6).
16"-P-1004-B3	29.5 / 40.0	R-101 PSV set 36 + P-101 shut-off rise 4 bar (Flexibility tab, assumed)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	15	largest of the three HMB cases. Header before the split: the full pump flow in every case.
16"-P-1005-B3	29.5 / 40.0	R-101 PSV set 36 + P-101 shut-off rise 4 bar (Flexibility tab, assumed)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	20	all of P-101 (FV-104 ≥ 77 % stroke, model 1.3·x split). Rated at the full pump flow: FV-104 can send all of it through E-101 (from 77 % stroke).
16"-P-1006-B3	29.5 / 40.0	R-101 PSV set 36 + P-101 shut-off rise 4 bar (Flexibility tab, assumed)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	40	by-pass rejoins: full P-101 flow. Return header after the by-pass rejoins: full pump flow.
16"-P-1007-B3	29.5 / 40.0	R-101 PSV set 36 + P-101 shut-off rise 4 bar (Flexibility tab, assumed)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	20	FV-104 at 0 %: all of P-101 by-passes E-101. By-pass flow falls toward zero as FV-104 opens: keep a flushing connection; slurry settles in a stagnant by-pass.
4"-P-1008-B3	30.4 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	H	30	largest of the three HMB cases. Overflow driven only by ΔP R-101 → R-102 (0.4 bar) + static head: keep it short and self-draining.
8"-P-1009-B1	30.4 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	59 / 250	loss of E-103 cooling: R-101 overhead temperature → R-101 design (SL_ASSUMED.Tdes1)	B1	—	25	largest of the three HMB cases. Gravity return at loop pressure: no pocket, slope to R-101.
4"-P-1010-B3	29.9 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	160 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	15	largest of the three HMB cases
3"-P-1011-B3	32.9 / 40.0	R-102 PSV set 36 + P-102 differential 3 bar (model K.pumpDP), rounded up	160 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B3	PP	20	largest of the three HMB cases
3"-P-1012-B1	29.4 / 40.0	R-102 PSV set 36 + P-102 differential 3 bar (model K.pumpDP), rounded up	160 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B1	H	40	largest of the three HMB cases. Clear liquid after the candle filter.
4"-P-1013-B1	29.0 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	45 / 80	max operating 45 °C + 30 °C, rounded up (typical)	B1	—	20	largest of the three HMB cases. Spec break at LV-104 (B1 → A2).
4"-P-1014-A2	2.0 / 10.0	product line to the atmospheric tanks, 10 bar g (assumed)	45 / 80	max operating 45 °C + 30 °C, rounded up (typical)	A2	—	300	largest of the three HMB cases. Sees loop gas if V-104 loses its liquid seal (EQ-ep-12).
10"-PG-2001-B2	30.4 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	200 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B2	—	20	largest of the three HMB cases. Vapour line to the condenser: no insulation needed for heat; self-draining to E-103.
8"-PG-2002-B2	30.4 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	59 / 250	loss of E-103 cooling: R-101 overhead temperature → R-101 design (SL_ASSUMED.Tdes1)	B2	—	15	largest of the three HMB cases. Two-phase: slope to V-103, no pockets.

Line number	P op / des bar g	Design pressure source	T op / des °C	Design temperature source	Spec	Ins.	Le m	Design-flow basis and notes
6"-PG-2003-B2	29.9 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	59 / 250	loss of E-103 cooling: R-101 overhead temperature → R-101 design (SL_ASSUMED.Tdes1)	B2	—	40	largest of the three HMB cases
8"-PG-2004-B2	29.9 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	160 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B2	H	30	largest of the three HMB cases. Gas + saturated C_6 vapour: insulate to avoid condensation slugs ahead of the trickle bed.
8"-PG-2005-B2	29.4 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	159 / 260	R-103 design temperature (SL_ASSUMED.Tdes3); loss of E-106 cooling	B2	PP	20	largest of the three HMB cases. Two-phase (flash estimate with the model's PsatC).
8"-PG-2006-B2	29.0 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	45 / 260	R-103 design temperature (SL_ASSUMED.Tdes3); loss of E-106 cooling	B2	—	15	largest of the three HMB cases. Two-phase into V-104 inlet device.
8"-H-2007-B2	29.0 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	45 / 80	max operating 45 °C + 30 °C, rounded up (typical)	B2	—	30	largest of the three HMB cases. Slope back to V-104; liquid knock-out ahead of K-101 (LSHH-104 protects).
8"-H-2008-B2	31.3 / 42.0	K-101 discharge design = PSV-112 set (SL_ASSUMED.PdesK)	60 / 90	max operating 60 °C + 30 °C, rounded up (typical)	B2	—	30	largest of the three HMB cases. Spec break at FV-103 (42 → 36 bar g design); check valve NRV-112 (SL-R22).
8"-H-2009-B2	31.3 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	60 / 250	R-101 design temperature (SL_ASSUMED.Tdes1)	B2	—	30	largest of the three HMB cases. Enters R-101 below the liquid: design temperature of R-101.
6"-H-2010-B1	35.0 / 40.0	supply 35 bar g (model K.Pmu) + margin; header design outside the unit (assumed)	40 / 70	max operating 40 °C + 30 °C, rounded up (typical)	B1	—	150	largest of the three HMB cases. Spec break at PV-105 into the loop.
1-1/2"-H-2011-B2	29.0 / 36.0	loop design pressure = PSV-101/102/104 set (SL_ASSUMED.Pdes)	45 / 80	max operating 45 °C + 30 °C, rounded up (typical)	B2	—	20	largest of the three HMB cases
3"-H-2012-A2	5.0 / 10.0	fuel-gas header 5 bar g (model K.Pfuel) → 10 bar g (assumed)	45 / 80	max operating 45 °C + 30 °C, rounded up (typical)	A2	—	150	largest of the three HMB cases. HIC-107 is the pressure break: the line must take loop gas if HIC-107 is opened wide.
10"-S-3001-A1	4.5 / 8.0	D-101 PSV-109 set (SL_ASSUMED.Psv109)	156 / 180	saturation at the PSV-109 set	A1	H	100	largest of the three HMB cases
8"-S-3002-A1	10.0 / 12.0	MP steam header 10 bar g + 2 bar (assumed)	184 / 200	saturation at 12 bar g + margin	A1	H	100	model maximum K.msMax (start-up heating). Start-up only (HS-101 HEAT). The E-101 shell is the D-101 water side, protected at 8 bar g (EQ-ep-3).
6"-S-3003-A1	10.0 / 12.0	MP steam header 10 bar g + 2 bar (assumed)	184 / 200	saturation at 12 bar g + margin	A1	H	60	model maximum K.msj (TV-102 wide open)
3"-BFW-3004-A1	10.0 / 15.0	BFW header 10 bar g (model K.Pbfw) + pump shut-off margin (assumed)	105 / 140	operating + 30 °C (typical)	A1	H	60	largest of the three HMB cases
8"-BFW-3005-A1	4.5 / 8.0	D-101 PSV-109 set (SL_ASSUMED.Psv109)	156 / 180	saturation at the PSV-109 set	A1	H	15	circulation ratio 8 × steam (assumed; not in the model). Natural circulation (assumed): head = density difference downcomer − riser.
16"-S-3006-A1	4.5 / 8.0	D-101 PSV-109 set (SL_ASSUMED.Psv109)	156 / 180	saturation at the PSV-109 set	A1	H	15	circulation 8 × steam, quality 12.5 % (assumed). Steam–water riser (two-phase).
10"-CW-3007-A1	4.0 / 10.0	CW thermal relief set (SL_ASSUMED.Tsv)	32 / 65	CW supply, typical	A1	—	150	model K.Fcw3 (HV-113 open)
10"-CW-3008-A1	3.0 / 10.0	CW thermal relief set (SL_ASSUMED.Tsv)	51 / 90	max return 50.7 °C + 30 °C (typical)	A1	—	150	model K.Fcw3. Return above the typical 45 °C limit (EQ-ep-7).
8"-CW-3009-A1	4.0 / 10.0	CW thermal relief set (SL_ASSUMED.Tsv)	32 / 65	CW supply, typical	A1	—	150	econ() fixed CW term (60 kg/s); the model has no E-106 water flow

Line number	P op / des bar g	Design pressure source	T op / des °C	Design temperature source	Spec	Ins.	Le m	Design-flow basis and notes
8"-CW-3010-A1	3.0 / 10.0	CW thermal relief set (SL_ASSUMED.Tsv)	49 / 80	max return 48.7 °C + 30 °C (typical)	A1	—	150	econ() fixed CW term
6"-CW-3011-A1	4.0 / 12.0	jacket also takes MP steam: MP design 12 bar g up to the change-over valves (EQ-ep-4)	32 / 200	MP steam saturation (change-over jacket)	A1	—	80	model K.Fcwj (TV-102 wide open in COOL). Change-over jacket: the CW lines up to the block valves see MP steam.
16"-N-3012-A1	8.0 / 10.0	N_2 header 8 bar g + 2 bar (assumed)	30 / 65	ambient maximum, typical	A1	—	100	model HV-120 flow into an empty loop: $0.35\cdot\sqrt{(9 - 1.013)}$ kmol/s. Size from the model flow is unrealistic (EQ-ep-6). Check valve + double block at the loop tie-in.
8"-FL-3013-A2	0.5 / 3.5	flare header design 3.5 bar g (typical, assumed)	45 / 250	hottest loop gas (EDV location not fixed): R-101 design	A2	—	150	model EDV flow $0.12\cdot\sqrt{(P - 1.2)}$ kmol/s at the loop PSV set (36 bar g). Sized on Mach number at the assumed back pressure; API 521 back-pressure check in the Relief & vent tab.
12"-FL-3014-A2	0.5 / 3.5	flare header design 3.5 bar g (typical, assumed)	45 / 250	hottest loop gas: R-101 design	A2	—	150	model vent flow $0.3\cdot\sqrt{(P - 1.2)}$ kmol/s at the loop PSV set. Hand vent: a restriction orifice would cut the line size.

Insulation (typical): H heat conservation, PP personnel protection, — none. Le = assumed equivalent length including fittings.

4 · Engineering queries

Inconsistencies found while writing this line list and the heat and material balance. They are collected in the Project & deliverables document register (CHX-V1-PM-DOC-001).

ID	Query	Finding	Documents	Proposed resolution
EQ-ep-1	PFD has no stream for the FV-104 by-pass and the D-101 blowdown	P-101 always moves 606 t/h; at design 248 t/h of it by-passes E-101 through the three-way FV-104 (stream 10B here), and D-101 blows down 1 % of the steam (BD). PFD stream 10 shows only the 357 t/h through E-101, so the PFD alone under-states the pump and the return line by 70 %.	CHX-V1-PFD-001, CHX-V1-PE-HMB-001, P-101 datasheet	Add 10B and BD to PFD rev 1; P-101 and lines 16"-P-1004-B3, 16"-P-1006-B3 rated at the full pump flow.
EQ-ep-2	P-101 differential pressure has two values	The page's streamRows puts the pump-around discharge at PI-101 + 2 bar; the Flexibility tab illustration uses 3 bar at 606 t/h (shut-off 4.0). The hydraulic budget in part 6 needs the lines + E-101 + FV-104 to fit inside the rated differential.	ui2.js streamRows, Flexibility tab, CHX-V1-PE-LL-001, P-101 datasheet	Fix the rated differential on the P-101 datasheet from part 6 (E-101 allowable ΔP set from the budget); align streamRows and the Flexibility illustration (lead).
EQ-ep-3	MP steam reaches the E-101 shell protected at 8 bar g	HS-101 HEAT puts MP steam (10 bar g) on the E-101 shell, which is the D-101 water side protected by PSV-109 at 8 bar g (SL_ASSUMED). Line 8"-S-3002-A1 is designed for 12 bar g but the equipment it feeds is not.	CHX-V1-PE-LL-001, Safety layers SL-R12, CHX-V1-PE-RLB-001, D-101/E-101 datasheets	Either D-101/E-101 shell design ≥ 12 bar g with PSV-109 re-set, or a pressure-reducing station + PSV on 8"-S-3002-A1 (Relief & vent tab).
EQ-ep-4	R-102 jacket: steam and cooling water on the same jacket	The jacket takes MP steam (10 bar g, 184 °C) in HEAT and CW in COOL. TSV-102J is set at 10 bar g = the MP steam pressure (relief at operating pressure), and the CW lines up to the change-over valves see steam.	Safety layers SL-R15, CHX-V1-PE-LL-001 6"-CW-3011-A1, R-102 datasheet	Jacket and CW branch up to the block valves designed 12 bar g / 200 °C; TSV-102J set ≥ 12 bar g; double block and bleed at the HS-102 change-over.
EQ-ep-5	TSV-102F set close to the F-102 operating pressure	F-102A/B sit on the P-102 discharge (stream 14 32.6 bar g at design, 32.9 at 120 %). TSV-102F is set at the loop design 36 bar g, so the operating pressure is 91 % of the set (typical limit 90 % for spring valves).	Safety layers SL-R16, CHX-V1-PE-LL-001 3"-P-1011-B3, F-102 datasheet	F-102 design 40 bar g (= the pump discharge side, as 3"-P-1011-B3) and TSV-102F set 40 bar g.
EQ-ep-6	Nitrogen flow in the model is far above a plant N_2 supply	HV-120 delivers $0.35\cdot\sqrt{(9 - P)}$ kmol/s: 79,814 Nm³/h into an empty loop, which makes 16"-N-3012-A1. A typical purge supply is a few thousand Nm³/h (assumed), and nothing stops loop gas flowing back into the 8 bar g header.	model.js (Qn2), CHX-V1-PE-LL-001, P&ID CHX-V1-PID-001	Define the N_2 header capacity, add a restriction orifice RO-120 + check valve and double block at the tie-in; cap the model N_2 flow (lead, model change).
EQ-ep-7	Cooling-water return temperatures above typical practice	E-103 returns 47.7 °C at K.Fcw3 = 120 kg/s and E-106 45.9 °C at the econ() 60 kg/s (design); at 120 % E-106 returns 48.7 °C. Typical maximum return 45 °C (fouling, scaling). The model has no E-106 water flow.	model.js K.Fcw3, econ(), CHX-V1-PE-HMB-001, E-103/E-106 datasheets	E-103 CW ≥ 145 kg/s and E-106 ≥ 77 kg/s for a 45 °C return; put the E-106 flow on its datasheet.
EQ-ep-8	Recycle gas carries no cyclohexane vapour in the model	At V-104 (44.9 °C, 30.0 bar a) the model's own PsatC gives 1.1 mol % cyclohexane in the gas ≈ 1.84 t/h circulating through K-101 and 81 kg/h lost to fuel gas; the HMB follows the model (dry gas).	model.js, CHX-V1-PE-HMB-001, K-101 datasheet	K-101 datasheet with the saturated gas (molar mass and flow); product loss to fuel gas in the economics.
EQ-ep-9	FV-103 at 92 % open in the 120 % case	The design-basis maximum rate needs FV-103 at 92.1 % (design 50 %, 70 % case 27 %); typical practice keeps a control valve ≤ 80–90 % open at maximum flow. The Operating-windows tab finds FV-103 as the top of the window.	CHX-V1-PE-HMB-001, Operating windows / Flexibility tabs, FV-103 datasheet	Size FV-103 for the 120 % case at ≤ 80 % (larger Cv) together with K-101 head (Flexibility recommendation).
EQ-ep-10	Power in econ() differs from the hydraulic estimates	econ() books K-101 1200 kW, P-101 160 kW, P-100 90 kW, P-102 30 kW; part 6 estimates the absorbed power from flow × head (see the table).	model.js econ(), CHX-V1-EE-LL-001	Electrical load list from the equipment datasheets (absorbed power + motor margin); keep econ() as a training price model and say so.
EQ-ep-11	Hot-hydrogen material for spec B2	B2 lines run at up to 200 °C (design 250–260 °C) with a hydrogen partial pressure ≈ 14 bar. Carbon steel in hot hydrogen needs an API 941 (Nelson curve) check.	CHX-V1-PE-LL-001 spec B2, CHX-V1-ME-EQL-001	Mechanical to confirm B2 material (killed CS + PWHT or Cr-Mo) against API 941 at the design temperature.
EQ-ep-12	Product line downstream of LV-104 and V-104 design temperature	4"-P-1014-A2 (class 150, 10 bar g assumed) sees loop gas at 36 bar g if V-104 loses its liquid seal (Safety layers SL-R6 recommends LSLL-104 + XV-104). On loss of E-106 cooling the V-104 inlet reaches the R-103 temperature (159 °C) while its outlets are designed for operating + 30 °C.	CHX-V1-PE-LL-001, Safety layers SL-R6, CHX-V1-PE-RLB-001, V-104 datasheet	Gas blow-by relief case on the product line/tanks (Relief & vent tab) or spec B1 up to a second isolation; V-104 design temperature to cover loss of E-106 cooling.

ID	Query	Finding	Documents	Proposed resolution
EQ-ep-13	Catalyst slurry below 1 m/s at the lowest case	16"-P-1005-B3 0.90 m/s, 16"-P-1007-B3 0.61 m/s, 4"-P-1010-B3 0.77 m/s: the E-101 branch and the FV-104 by-pass swap flow as FV-104 moves, and the P-102 suction cannot hold 1–2 m/s over a 1.7 : 1 flow range with standard sizes (3" gives 2.30 m/s at the rated flow).	CHX-V1-PE-LL-001, P&ID CHX-V1-PID-001, P-101/P-102 datasheets	Flushing connections (cyclohexane) on the three lines; FV-104 minimum by-pass stop; accept 3" P-102 suction at the rated case or a minimum-flow recycle on P-102; settle the catalyst loading (not in the model).

5 · Assumptions

P-101 differential 2 bar (page streamRows), shut-off rise 4.0 bar (Flexibility tab illustration); P-100 shut-off from the model (no feed once PI-101 ≥ 36 bar a); D-101 natural-circulation ratio 8; E-106 cooling water 60 kg/s (econ() term); flare back pressure 1.5 bar a and k = 1.4 for the Mach number; N_2 and flare flows from the model's valve equations (EQ-ep-6). Piping classes, corrosion allowances, insulation and equivalent lengths are typical practice for teaching, not a project specification.